

Week-5 Speech Tech

L1 Hidden Markov Model

- HMM (hidden markov model)

↳ sequence of states is hidden

→ we assume, we can't directly observe today's weather (i.e. it stays hidden)

→ let's say instead observe a related phenomenon like dryness of grass.

→ for now the states of grass could be
- dry, dryish, damp, soggy.

		Dry	Dryish	damp	soggy
W	S	0.6	0.2	0.15	0.05
	G	0.25	0.25	0.25	0.25
	R	0.05	0.1	0.35	0.50

This is not transition probability, this is observational probability.

→ this matrix gives probab. distribuⁿ of obs. given the weather condition

→ it is discrete

- we have 2 pieces of info -

→ weather condiⁿ the prev. day
↳ through transition probab.

→ condition of the grass today
↳ observaⁿ probab.

- Now consider a scenario,

→ I know Sunday is → Sunny / C / R

→ grass obs. for next 5 days → DY, DY, S, Da, S

→ what is the most possible seq. of weather?

DY → Dary
S → Soggy
Da → Dampy

→ we use Viterbi algo.

→ Viterbi of HMM is almost same as Markov chain except that HMM has observation probabilities in addⁿ.

→ let's say

assume first day is cloudy

5 days grass → Dary, Dary, Soggy, Damp, S

e.g. C → S, S, C, S, S

$P(S, S, C, S, S \mid \text{grass 5 days})$

$= P(\text{grass 5 days} \mid S, S, C, S, S)$

$\times P(S, S, C, S, S)$

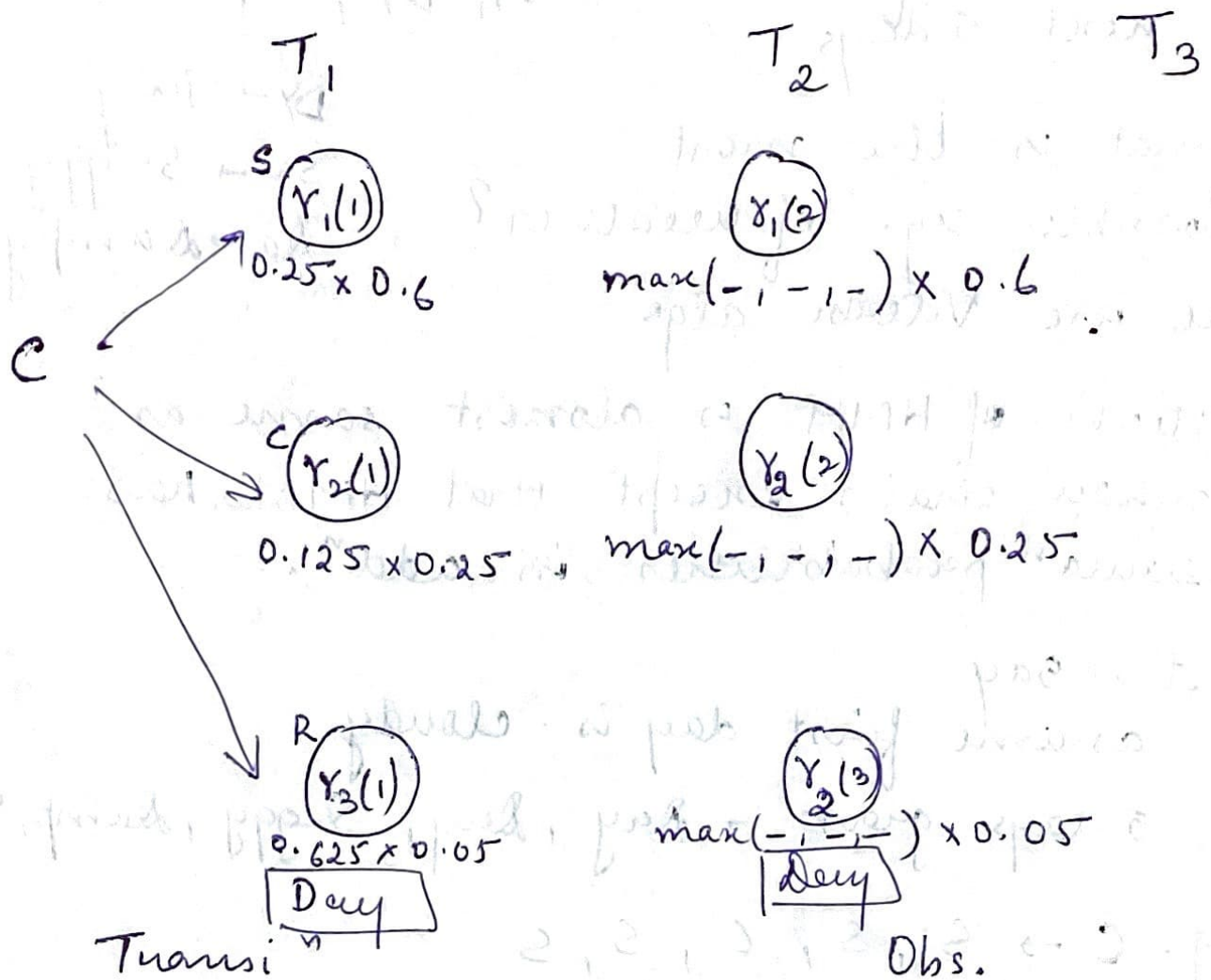
as per the Bayes' Rule → obs. probab.

$= [P(\text{Dary} \mid S) P(\text{Dary} \mid S) P(\text{Soggy} \mid C)$

transition $P(\text{damp} \mid S) P(\text{damp} \mid S)] \times$

probab. $[P(S \mid C) P(S \mid S) P(C \mid S) P(S \mid C)$
 $P(S \mid S)]$

→ but we will have $3^5 = 343$ paths, we just calculate just 1 path.



	S	C	R
S	0.5	0.375	0.125
C	0.25	0.125	0.625
R	0.25	0.075	0.675

same as before

Dey Deyin Damp so

S 0.6

C 0.25

R 0.05

get the max & then backtrack as done in Viterbi.

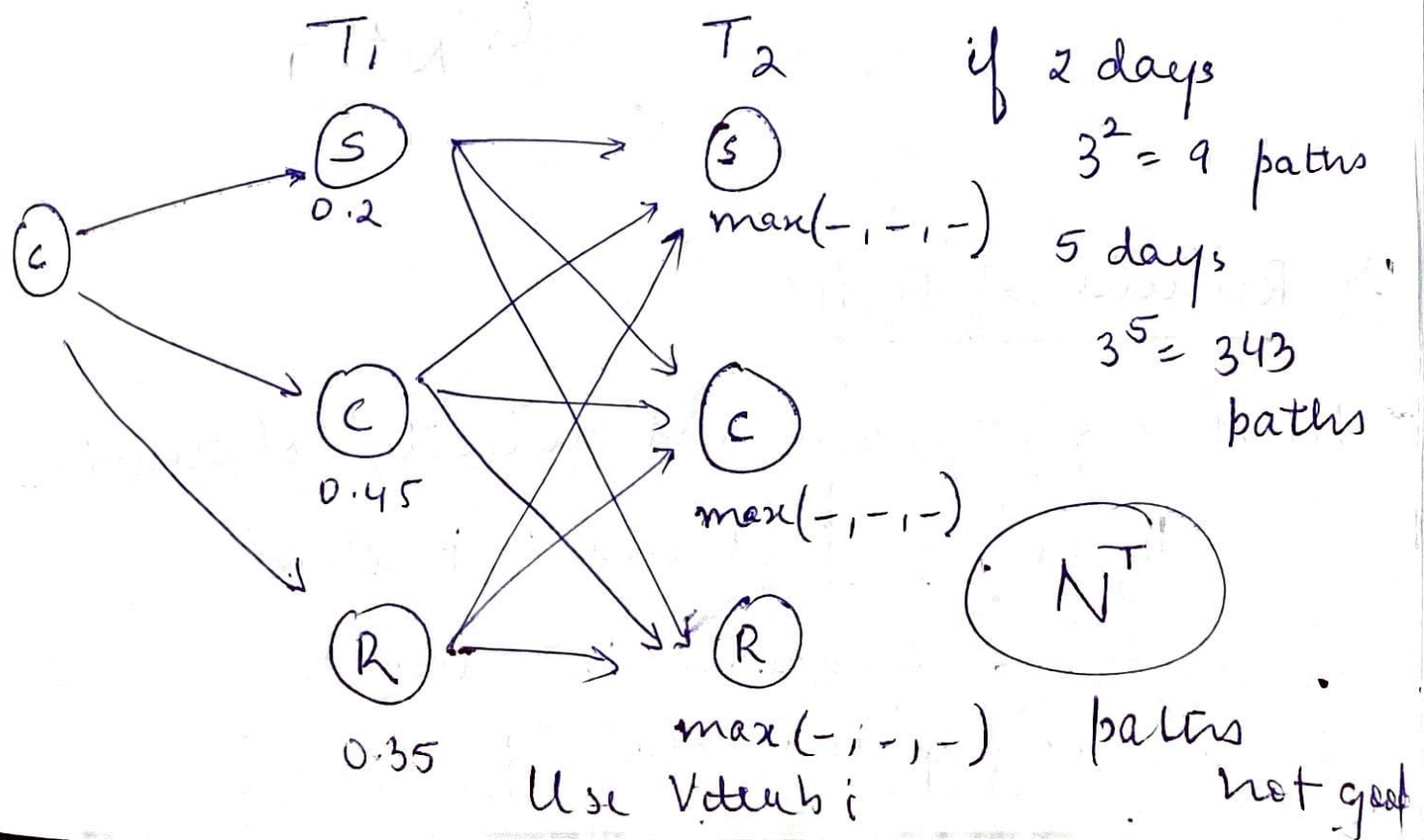
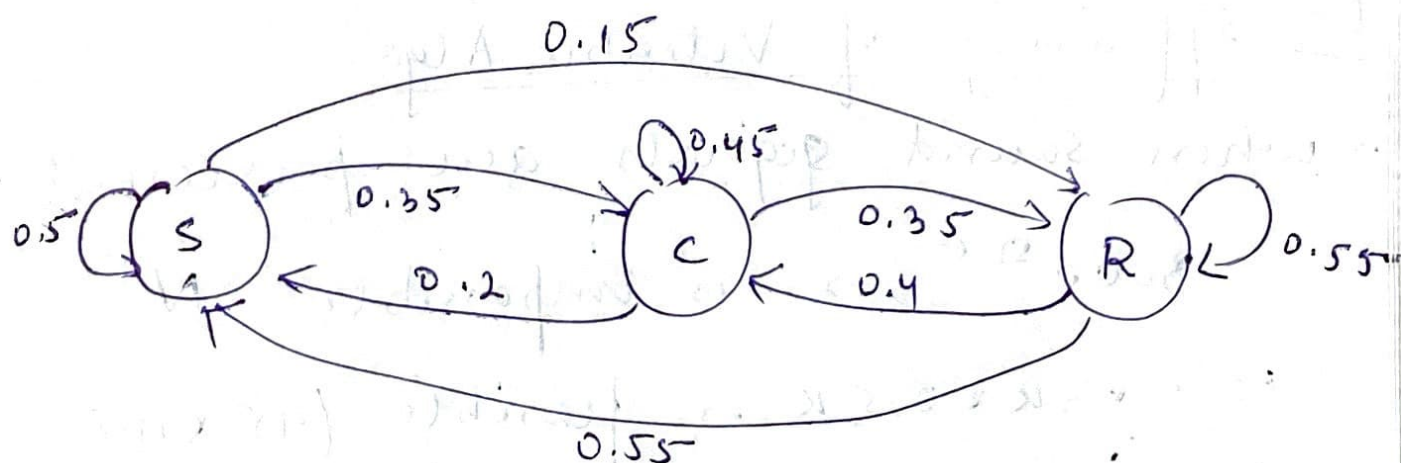
L2 Review of Weather Prediction Example

3 states $\rightarrow S, C, R$

Transition matrix

↳ that tells the probability of the given state today given what was the state yesterday.

3 states $\rightarrow 3 \times 3$ matrix



- if today is cloudy, what is probab. of next 2 days being S & S

$$P(S|C) \times P(S|S) = 0.1$$

- given today is cloudy, what is best forecast for T_1 & T_2 .

Use Viterbi $\Rightarrow$ efficient

only $N^2 \cdot T$ computat^{ns}

L3 Efficiency of Viterbi Algo.

- when sound signals are processed -

3000⁵⁰⁰ $\Rightarrow$ is impossible N^T

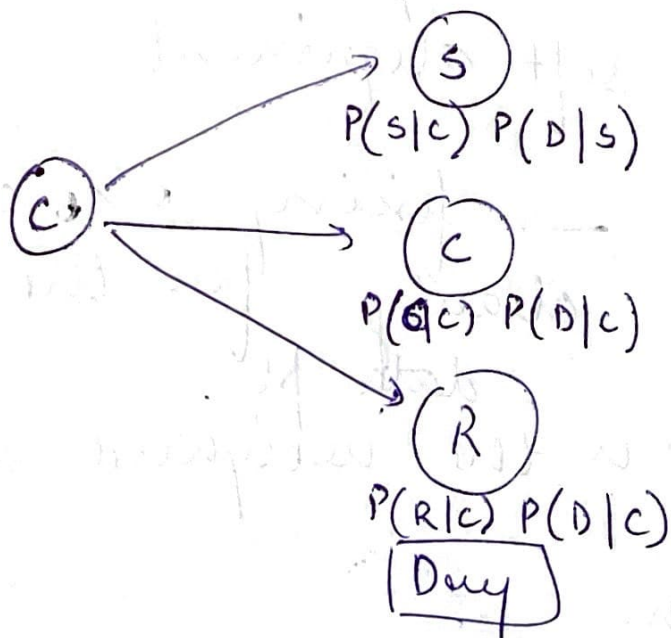
$3000 \times 3K \times 0.5K \Rightarrow$ feasible (45×10^8)
 $\hookrightarrow N^2 \cdot T$

L4 Review of HMM

- HMM $\rightarrow$ states are not directly observable

$b_j(0 \neq)$ } observable probab.

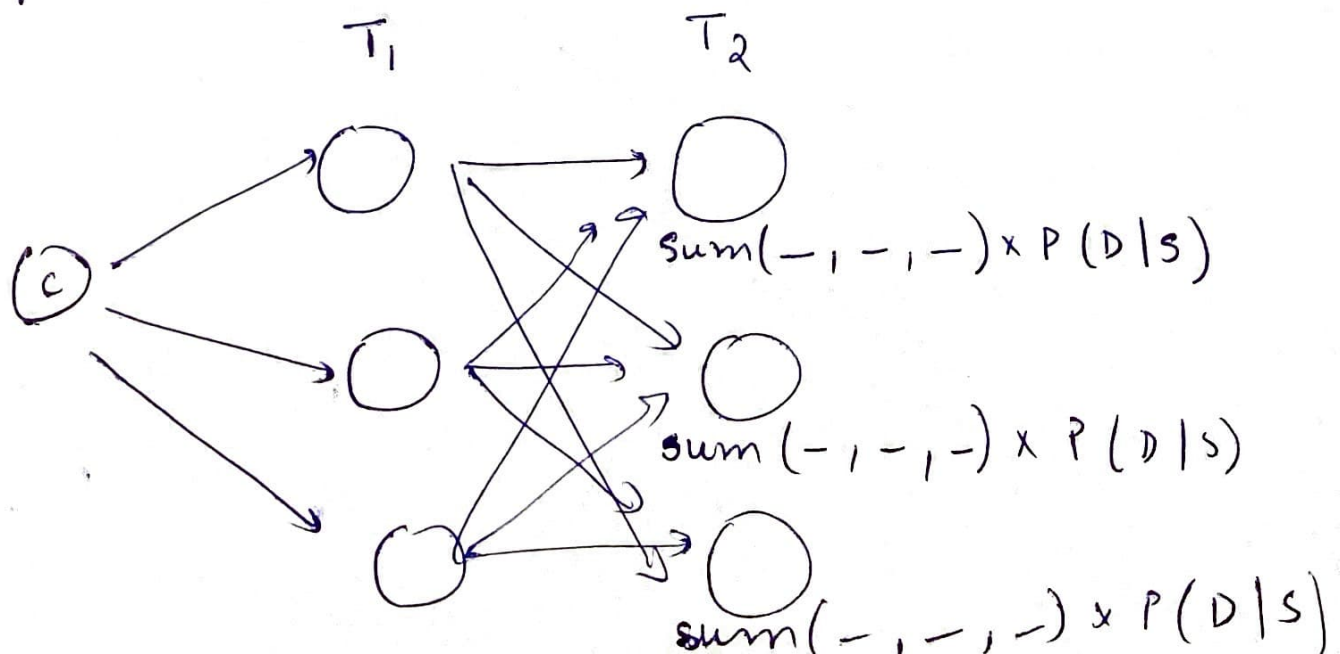
everything remain but we multiply the obs. probab.



multiply
it here

Forward Algo for HMM

- similar to Viterbi, but here we use sum instead of max.
- It answers questions like - how probable is today being sunny given C & D ?
- While Viterbi gives the best possible sequence.



Viterbi $\rightarrow$ hard alignment

Forward $\rightarrow$ soft alignment

- hard alignment $\rightarrow$ fixing one class clearly for the data pt.
- soft alignment $\rightarrow$ is the weighted avg.

$$p_1 \times h_1 + p_2 \times h_2$$

$$p_1 + p_2 \dots$$